

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
Student Name:	Aswini S	Reg. No.:	192511077
Date:		Duration:	60 minutes

Code of Conduct

I S Aswini (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: aswini

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. Understand the Problem Statement

- Explain the problem in your own words.
- Identify the objectives of the proposed system.
- State the scope of the problem.

2. Identify Key Stakeholders and Challenges

- Identify all the stakeholders involved in the system.
- Explain the roles and responsibilities of each stakeholder.
- Discuss the major challenges and issues faced by the stakeholders.

3. Analyze the Current Approach

- Describe the existing system or current process.
- Identify its strengths and limitations.
- Analyze the problems or gaps that need to be addressed.

4. Propose a Suitable Solution Using Software Engineering Principles

- Design a software-based solution for the given problem.
- Describe the major functional modules of the proposed system.
- Identify the functional and non-functional requirements.
- Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

5. Justify the Chosen Methodology/Framework

- Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).

- Justify why the selected methodology/framework is suitable for the proposed system.
 - Explain how it supports successful project development and maintenance.
- 6. Discuss Expected Outcomes and Limitations**
- Describe the expected benefits of the proposed solution.
 - Explain how the solution addresses the identified challenges.
 - Discuss potential risks, implementation challenges, and limitations.
 - Suggest possible future enhancements.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/framework and provides	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing;

	strong technical justification.			report lacks organization and clarity.
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Criteria	Mark
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25

CASE STUDY ANALYSIS

Digital Platform to Crowdfund Societal Challenges

1.Understand the Problem Statement

The problem is to design a system where the citizen, communities, organizations, and other stakeholders report real-world societal challenges. They will be able to document the problem with a description, images/videos, location, and other relevant digital records. The system will help categorize and collate the information received about the challenges and enable the collaboration of universities, industries, experts, and stakeholders to find possible solutions.

Objectives of the Proposed System

The objectives of the proposed system include:

- Provide a platform to report a society-related problem
- Allow the users to document the problems with text, images, video, location, and other relevant information
- Categorize the problems and collate the information for easier tracking
- Ensure that problems are reported to the relevant stakeholders, including universities, industries, experts, and other solution providers
- Enable stakeholder collaboration and coordination
- Track the status and progress of the reported issues
- Provide dashboards and reports to monitor and manage the processes
- Improve transparency, reporting, and problem-solving process

Scope of the Problem

The scope of the problems includes the end-to-end process of societal problem reporting, management, and stakeholder collaboration. The system will be used by the citizen, communities, NGOs, local governments, administrators, industries, universities, experts, and other stakeholders. The following are the main scope areas:

- User registration and authentication
- Societal problem reporting
- Image and video upload
- Location identification and tagging
- Problem categorization
- Search and filtering options
- Stakeholder collaboration and coordination
- Status tracking and monitoring
- Notifications and reminders
- Dashboard and reports
- Database and access rights

2.Identify Key Stakeholders and Challenges

Key Stakeholders and Their Roles

- **Citizens / Problem Submitters**
 - Report societal problems through the platform.
 - Provide descriptions, photos, videos, location, and documents.
 - Track the progress of submitted problems.
- **Community Organizations / NGOs**
 - Identify local community challenges.
 - Submit and verify relevant problems.
 - Participate in finding suitable solutions.
- **Universities / Educational Institutions**
 - Identify problems suitable for student or research projects.
 - Provide technical knowledge and innovative solutions.
 - Collaborate with other stakeholders.
- **Industry Partners**
 - Provide technical expertise, resources, and practical solutions.
 - Collaborate with universities and communities.
 - Support implementation of suitable solutions.
- **Experts / Solution Providers**
 - Analyze submitted challenges.
 - Suggest technical or domain-specific solutions.
 - Provide guidance during implementation.
- **Government / Local Authorities**
 - Review challenges related to public services.
 - Coordinate with relevant departments.
 - Monitor and support the resolution of suitable problems.
- **System Administrator**
 - Manage users and platform access.
 - Verify and manage submitted challenges.
 - Monitor system activities and generate reports.
 - Maintain system security and performance.

Major Challenges Faced by Stakeholders

- **Citizens:** Difficulty finding the correct organization to report a problem to.
- **NGOs / Communities:** Difficulty coordinating multiple stakeholders.
- **Universities:** Difficulty identifying real-world problems suitable for projects.
- **Industries:** Difficulty finding relevant societal challenges where their expertise can be applied.
- **Experts:** Difficulty accessing organized and reliable problem information.
- **Government Authorities:** Large numbers of problems can make monitoring and prioritization difficult.
- **Administrators:** Need to manage large amounts of data while maintaining **security, accuracy, and system reliability**.

3. Analyzing the Existing Approaches

Existing system / current process

- Societal issues are raised by the local authorities, in social media, on websites, through helpline numbers, or through direct conversation.

- There may be different data banks with different organizations.
- The problem is reported manually over calls, messages, or through documentation.
- There may not be a collaborative portal that connects the citizens, academia, industry professional or experts together.

Strengths

- There should be a procedure for the local authorities to report immediate issues.
- The problems can be directly contacted by the local authorities.
- The issues can be raised through social media and gain popularity.
- Various institutions can contribute their expertise to solve the problems.

Limitations

- There is no digital central database for all societal issues.
- The problems are not prioritized.
- Similar problems are not grouped together.
- There is no direct connection between the person reporting the issue and the solving entity.
- It may not be easy to track the progress of a problem.
- The information management may become very manual.
- There is no collaboration and communication framework in place.
- There is no dashboard or portal to track these issues.
- There is no proper storage mechanism and access control system.

Problems / gaps

- A centralized digital database is needed.
- Prioritization framework is required.
- There is a need to have a system that connects the citizens to appropriate academia, industry, expert, or organization to solve the problems.
- The process and tools for tracking the progress of a problem should be developed.
- Communication and collaboration framework should be designed.
- Dashboards and reports to monitor societal issues should be produced.
- The database should be properly stored with a suitable access control system.

4. Propose a suitable solution using software engineering principles

Software Based Solution

A centralized web-based digital platform for reporting and managing societal challenges will be designed and implemented. The platform will enable citizen and organizations to register, login and report societal problems. Users will upload information related to the problems including description, pictures, videos, location and documents. The reported problems will be categorized to aid management. Organizations and individuals including universities and industries can search for problems and collaborate to address them. An administrator will approve the reported problems and deal with other aspects of problem management. The users will track the progress of their submitted problems. Major modules

include: user management, problem submission, problem categorization, search, collaboration, progress tracking, notifications, and administration, reports and dashboard.

Functionality

The online societal problem management system will have the following functionalities:

- User registration
- Problem submission with description and file upload.
- Image, video, and document management.
- Location information management.
- Problem categorization.
- Searching and filtering of problems.
- Collaboration with stakeholders.
- Status and progress tracking of problems.
- Notifications and Alerts.
- Administrator functions.
- Report and dashboard generation.
- Non-Functionality
- Security of information
- Easy to use
- Reliability
- Performance
- Scalability
- Maintainability
- Availability

Software Engineering Principles

- Modularity – The application will be split into distinct modules like users, problem, collaboration and administration.
- Scalability – database and application architecture should be designed in a way that more users and data can be accommodated.
- Maintainability – Code should be clean and proper documentation provided. Decoupling the application into sections such as front-end, back-end and database will enhance easy maintenance.
- Reliability – The application will have proper error handling and backup procedures to enhance reliability.
- Usability – The application will have a friendly interface and navigation to enable end-users to perform actions without confusion.
- Security – The application will have safe and reliable file upload and management solutions to prevent attack and unauthorized access. Users will be provided with login credentials to access the application and in case of a security breach, the information will be safe since it is encrypted.

5. Justification of the Chosen Methodology/Framework

Proposed Methodology: Agile – Scrum

- Agile Scrum methodology is selected for developing the proposed Digital Platform for Crowdsourcing Societal Challenges.
- The system has multiple modules including user management, problem submission, categorization, collaboration, notifications, dashboards, and administration. The project can be divided into smaller sprints with each sprint targeting to develop and test a set of features.

Justification

- Incremental development: The system can be developed module by module instead of building everything from scratch
- Feedback: Feedback from users, administrators, and stakeholders can be incorporated after each iteration
- Flexibility: New features or changes can be easily incorporated in later iterations
- Testing: Each module can be tested individually before integrating into the whole system
- Risk reduction: Potential issues can be detected and resolved during the initial phases
- Collaboration: The project requires close collaboration between developers, testers, designers, and stakeholders

Example Sprint Plan

- Sprint 1: User registration, login and role management
- Sprint 2: Societal problem submission and file upload
- Sprint 3: Problem categorization, search and filtering
- Sprint 4: Collaboration and solutions management
- Sprint 5: Notifications, dashboards and progress tracking
- Sprint 6: Integration, security and performance testing, and deployment
- Support for successful development and maintenance
- Supports Continuous Testing and Improvement: The approach allows for continuous testing and improvement of the system throughout development
- Facilitates early defect detection and resolution: This methodology allows for early defect detection and resolution
- Enables stakeholder feedback and system tweaking: The approach enables the project team to solicit feedback from stakeholders and make necessary improvements
- Supports iterative prototyping: The approach allows for the development of new modules and features without having to rebuild the whole system
- Ensures system reliability, scalability, and security: The approach supports system reliability, scalability, and security since it allows for regular testing
- Supports future system improvements: After deployment, system maintenance and improvement can be done using new Scrum sprints.

6. Discuss Expected Outcomes and Limitations

Expected Benefits

- Offers a single source for reporting societal issues
- Enables easier submission of problems by citizens and organizations
- Facilitates collaboration between citizens, universities, industries, experts, and organizations
- Ensures categorized and trackable challenges

- Offers better visibility of the problems and their progress
- Eliminates the need for multiple communication channels
- Enables the administrators to monitor the challenges through dashboards and reports
- Improves transparency and stakeholder collaboration
- How the Solution Addresses the Challenges
- Scattered information
- A centralized online database
- Problem identification challenges
- Categorization and searchability
- Collaboration challenges
- A collaboration module where all stakeholders can connect
- Progress monitoring challenges
- Status and progress tracking system
- Manual monitoring of challenges
- Administrators' dashboard and automated alerts
- High volumes of problems and monitoring
- Database with searchability and filtering options
- Potential Risks and Implementation Challenges
- Data security risks due to the collection of users' data and documents
- Inaccuracies and duplication of problems reported
- Too many users causing system instability
- Varied technical fluency among users
- Integration with existing systems and tools, including location positioning and notifications
- Ongoing maintenance and server costs
- Need for a stable internet connection to access the platform

Limitations

- The system cannot guarantee solutions to all problems reported
- The quality of solutions depends on the involvement and commitment of all stakeholders
- Inaccuracies in information provided by the users need manual moderation
- The categorization and prioritization of problems might be computationally intensive
- Authorization and authentication of certain stakeholder groups, particularly organizations, could be an issue

Future Enhancements

- Design and develop a mobile application version for iOS and Android
- Advanced artificial intelligence for automated categorization, prioritization, and recommendations
- Multilingualism, including localized languages
- Real-time communication and collaboration tools
- Internet of things (IoT) enabled automated problem detection and monitoring
- Advanced data visualization and analytics
- Location-based notifications and recommendation system